Predicting Loan Default Risk

Model Development & Evaluation

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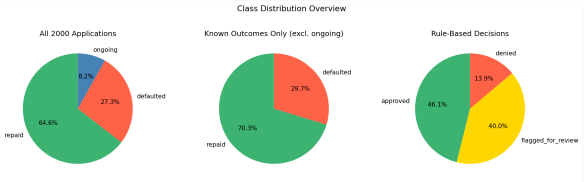
Exploratory Data Analysis

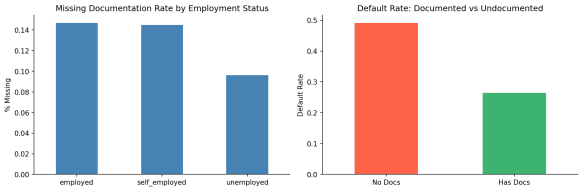
Target Imbalance & Data Scope

- Severe class imbalance in known outcomes (~85% Repaid, ~15% Defaulted) standard accuracy is misleading; F1 and AUC are the right metrics.
- ~8% of applications are `ongoing` no ground-truth label available, excluded from training.
- Statistical tests confirm ongoing apps are distributed randomly; exclusion introduces minimal selection bias.



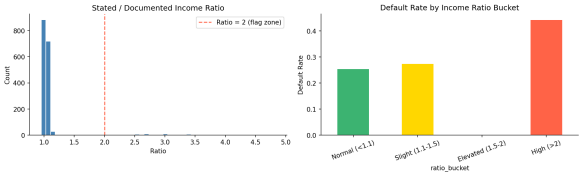
Missing Data as a Predictive Signal

- 15% of applications are missing `documented_monthly_income`.
- Missingness is not random; undocumented applicants default at a significantly higher rate.
- Missing documentation is a risk signal in its own right not merely a data quality issue.



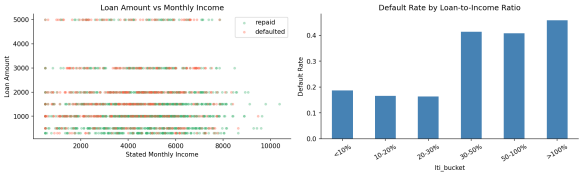
Behaviour Signals Over Raw Values

- When stated income is more than 2× documented income, default rates spike. Misrepresentation is a strong risk signal.
- Default rate jumps from ~17% to ~42% once the loan is more than 30% of monthly income.
- Raw dollar amounts matter less than how income *relates* to the loan size.



Loan Size Relative to Income is the Key Signal

- Default rate stays flat below 30% LTI, then jumps sharply above it.
- Above 30%, the applicant is likely stretched thin, even if their raw income looks fine.
- This is why `loan_to_income` becomes one of the strongest features in the model.

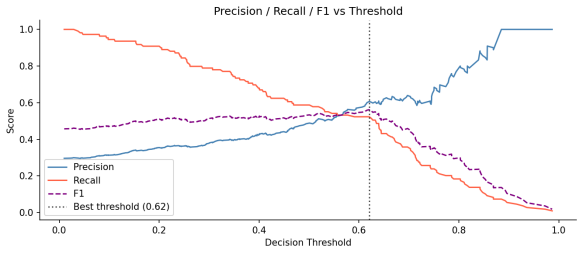


Model Choice

Why XGBoost?

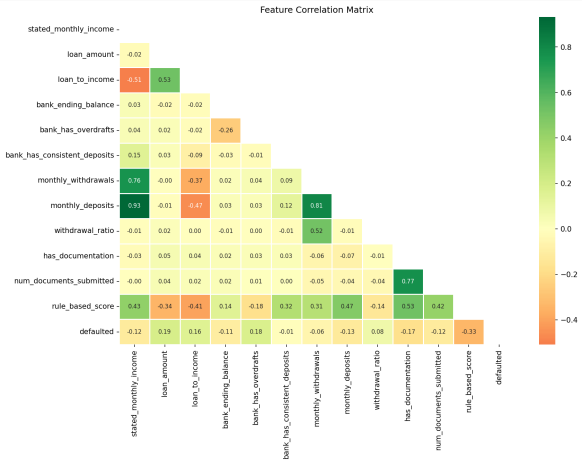
- Picks up non-linear interactions between features logistic regression can't do this.
- `scale_pos_weight` deals with class imbalance natively, no resampling needed.
- Threshold tuned for F1 on defaults, not the naive 0.5 cutoff.

5-fold CV AUC-ROC	0.683 ± 0.003
Train size	1,468 applicants
Test size	368 applicants



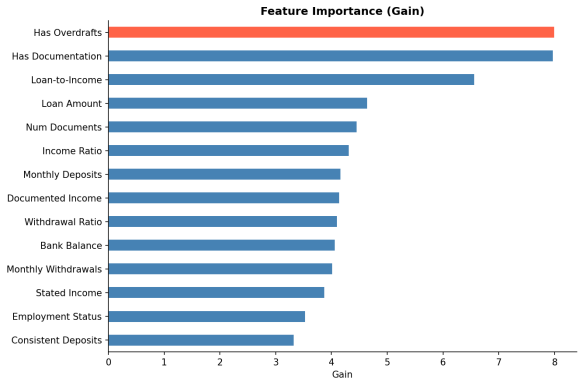
Why the Rule-Based Score Isn't Enough

- Overdrafts and missing documentation are the strongest correlates with default.
- Bank balance and consistent deposits are negatively correlated, as expected.
- The rule-based score has weak correlation with actual outcomes. This is the gap the model is built to close.



What Drives Default Risk?

- Gain-based importance pulled straight from the booster.
- `loan_to_income` and `withdrawal_ratio` top the list ratios beat raw numbers, same story as EDA.
- `has_documentation` in top 5 not having docs is a red flag on its own.



Explaining Individual Decisions

- Global importance $\neq$ why this applicant got flagged.
- Each denial shows the top features and actual values for that person.
- Regulator asks why someone was denied? The explainer answers it automatically.

Applicant 1185 FLAG 80.3% default probability

Feature	Value	Gain
Has Overdrafts	0	8.0
Has Documentation	0	8.0
Loan-to-Income	1.133	6.6
Loan Amount	2000	4.6
Num Documents	0	4.5

No documentation, no deposits, loan is 1.13$\times$ monthly income.
Each factor is visible and auditable.

Evaluation

What the Metrics Actually Mean

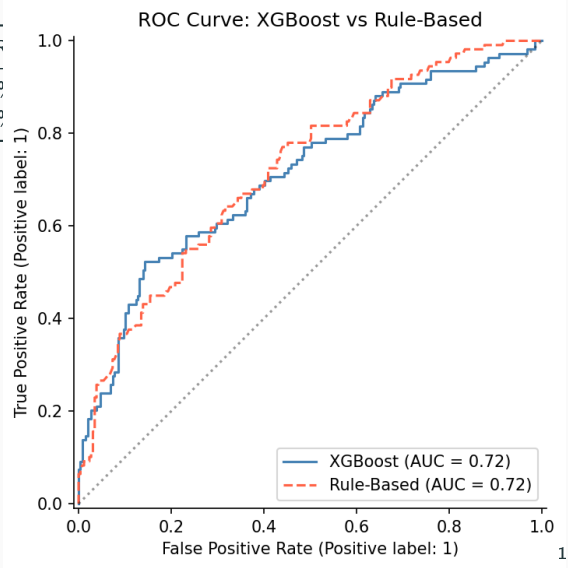
- **Precision:** of everyone flagged as a default, how many actually were? High precision means fewer good applicants wrongly denied.
- **Recall:** of all real defaults, how many did we catch? Low recall means defaults slipping through undetected.
- **F1:** balances precision and recall. More useful than accuracy when classes are imbalanced.
- **AUC-ROC:** ability to separate repayers from defaulters across all thresholds. 0.5 is random, 1.0 is perfect.
- **FPR** (False Positive Rate): good applicants wrongly denied. A business cost.
- **FNR** (False Negative Rate): real defaults approved by mistake. A direct financial loss.

XGBoost vs. Rule-Based Baseline

	Rec.	F1	AUC	FPR	FNR
XGBoost	67.0%	53.5%	70.9%	35.1%	33.0%
Rule	33.0%	43.1%	72.2%	8.5%	67.0%

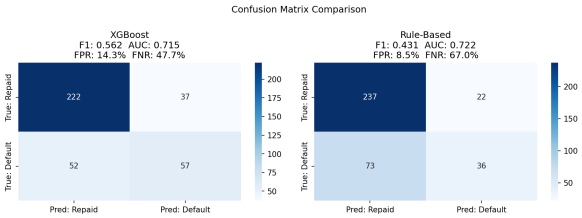
FNR drops from 67% to 33%. XGBoost catches twice as many defaults as the rule-based system.

Rule-based looks more precise only because it flags almost nobody. Threshold of 0.62 was tuned to maximise F1 on the default class.



The Deploy-Tomorrow Trade-off

- XGBoost catches **+37** more defaults per 368 applications vs. baseline (recall goes from 33% to 67%).
- Cost: **+69** good applicants wrongly denied (FPR goes from 8.5% to 35.1%).
- *Who decides?* The founders, weighing a missed default (full loan loss) against a lost customer (CAC). The threshold is a dial, not a fixed number.

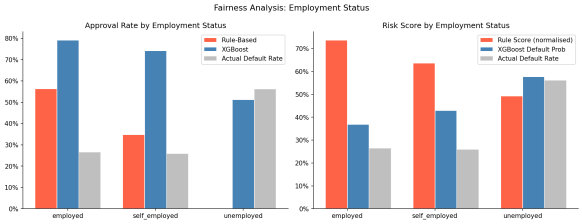


Fairness & AI Ethics

Where the Rule-Based System Fails on Fairness

Group	Default	Rule Appr.	ML Appr.
Employed	26.5%	56.3%	79.1%
Self-employed	25.9%	34.8%	74.1%
Unemployed	56.1%	0.0%	51.2%

- Employed vs. self-employed: nearly identical default rates, but the rule approves them at 56% vs. 35%. No risk justification for that 21-point gap.
- XGBoost brings it to 79% vs. 74%, much closer to the actual risk split.



The Harder Question: Should We Even Use This?

The case for the model

- Decisions are based on actual repayment behaviour, not assumptions about job type.
- Every denial is explainable with specific, per-applicant evidence.
- Catches twice as many defaults, which keeps the lender solvent and able to serve more customers.

The unresolved tensions

- `employment_status` is a proxy for protected characteristics. Even correct use carries disparate-impact risk.
- The model learned from historical loans. If past decisions were biased, it inherits that bias.
- An applicant denied by an algorithm has less recourse than one denied by a human. That matters.

The right answer is not "use the model" or "don't use it." It is to deploy it with ongoing audits, a human review layer for edge cases, and a clear process for applicants to contest decisions.

Production

If This Went Live Tomorrow. . .

- **Model drift:** a recession doubling default rates would silently degrade predictions unlike the rule-based system, which degrades *transparently* and *predictably*.
- **Distribution monitoring:** feature distributions (e.g., `loan_to_income` under inflation) must be tracked continuously; the static rule-based system requires no such infrastructure.
- **Regulatory audits:** handled the local explainer is deployed alongside the prediction API to generate automated, human-readable denial reasons per applicant.